Staff management strategy based on flight information to reduce queues at passport control.

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Abstract

Increasing congestion at airport border control has become a critical operational issue, particularly for non-EU passengers. With the ongoing rollout of the EU Entry/Exit System (EES), processing times are increasing, and long queues have already been observed at several European airports (e.g., Prague, Lisbon, and Milan Malpensa). This trend exacerbates queue lengths and passenger delays. This study proposes a data-driven decision-support framework that uses flight-level information to predict passenger flows and optimize staffing strategies at passport control. By combining predictive modeling with discrete-event simulation, the study evaluates different staffing and dynamic allocation policies. The results highlight how predictive approaches can significantly reduce waiting times while maintaining efficient use of border control resources.

1 Introduction

Airport border control systems operate under significant uncertainty due to fluctuating passenger arrivals driven by flight schedules. One of the most critical challenges is managing queues at passport control, especially for non-EU passengers who require manual document verification. Model of such process clearly observes the flow division for EU and non-EU passengers 1

This problem is expected to intensify with the introduction of the EU Entry/Exit System (EES), which increases processing time per passenger due to biometric registration and additional checks. As a result, traditional static staffing approaches are insufficient to handle peak demand efficiently. This study investigates how flight-level data can be used to improve staffing decisions. By leveraging predictive models and simulation techniques, the project aims to provide a structured approach to balancing service quality (waiting time) and operational efficiency.

2 Literature Review

Airport passenger flow management has been extensively studied in the context of increasing air traffic demand and operational complexity. Passenger arrivals at airports are inherently time-dependent and are strongly driven by flight schedules, leading to peak congestion periods and operational bottlenecks.

Queueing theory provides a fundamental framework for analyzing such systems. Classical queueing models, such as multi-server queues, are commonly used to represent service processes at airports. However, these models often rely on simplifying assumptions, such as stationary arrival rates, which do not accurately reflect real-world airport dynamics where passenger flows vary significantly over time.

To address these limitations, recent studies have increasingly adopted simulation-based approaches. Discrete-Event Simulation (DES) is particularly suitable for modeling airport systems, as it captures stochastic behavior, time-dependent arrivals, and resource constraints. In the context of air transport operations, [mdp19] propose a queue-based modeling approach that highlights how congestion emerges due to limited service capacity and variability in arrival processes. Their findings demonstrate the importance of accurately modeling demand fluctuations when designing service systems.

Simulation-based frameworks have also been widely applied to passenger flow analysis. For example, the simulator developed in [Kha23] models passenger movements through airport processes using a discrete-event approach, treating passengers as entities that pass through multiple service stages. The

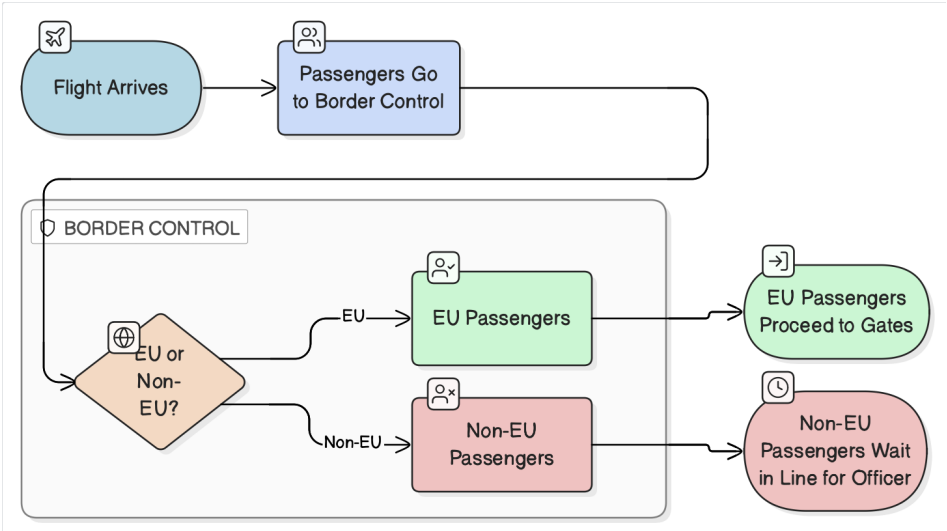

Figure 1: Model of passenger arrival business process.

study shows that passenger arrivals are closely linked to flight schedules and that service times can be effectively represented using probabilistic distributions. The results emphasize that adjusting service capacity dynamically can significantly reduce queue lengths and waiting times.

Furthermore, prior research indicates that static resource allocation policies are often insufficient in environments characterized by highly variable demand. Adaptive and dynamic staffing strategies have been shown to outperform static configurations, particularly in systems with time-dependent arrival patterns. Simulation-based decision-support tools enable the evaluation of such policies under realistic conditions, allowing managers to analyze trade-offs between service quality and operational efficiency.

Despite the extensive literature on queueing theory and simulation in airport operations, limited attention has been given to passport control systems under evolving regulatory constraints. The introduction of the European Union Entry/Exit System (EES) is expected to increase processing times due to biometric verification and additional checks, thereby intensifying congestion at border control, especially for non-EU passengers.

Moreover, few studies integrate flight-level demand modeling with dynamic staffing and resource allocation decisions. In practice, passenger arrivals occur in batches determined by flight schedules, making flight-level modeling essential for accurately capturing system dynamics.

This study addresses these gaps by combining flight-based passenger prediction with discrete-event simulation to evaluate staffing strategies at passport control. In particular, it focuses on dynamic allocation of border control officers to reduce waiting times while maintaining efficient resource utilization.

3 Research Questions

RQ1: To what extent can flight-based passenger flow prediction reduce waiting times at passport control by dynamical allocation of border control officers?

RQ2: How can simulation techniques be used to evaluate and compare staffing strategies under time-varying and uncertain passenger demand?

4 Methodology

The proposed framework is structured as a sequence of interconnected analytical layers, where each layer transforms operational flight information into progressively more decision-relevant outputs for staff management at passport control. The overall objective is to move from raw arrival information to staffing recommendations that reduce congestion and improve service performance for non-EU passenger flows.

4.1 Flight Information Layer

The first layer represents the operational characteristics of arriving flights that influence passenger demand at passport control. Each flight is described using variables such as scheduled arrival time, actual arrival time, aircraft type, aircraft capacity, load factor, route region, non-EU passenger share, and estimated walking time from gate to passport control. This layer provides the base information needed to estimate how many non-EU passengers each flight contributes to the border-control system and when they are likely to enter it.

Let:

- C_f denote the seat capacity of flight f,
- L_f denote the load factor of flight f,
- θ_f denote the proportion of non-EU passengers on flight f.

Then the estimated number of non-EU passengers associated with flight f is:

$$P_f = C_f \cdot L_f \cdot \theta_f \tag{1}$$

In implementation, this quantity is rounded to the nearest integer because passenger counts are discrete.

4.2 Passenger Flow Construction Layer

The second layer transforms flight-level observations into a time-indexed passenger demand series. Since passengers do not appear at passport control immediately after aircraft arrival, the framework adjusts each flight using estimated disembarkation and walking times. The resulting passenger arrivals are then aggregated into 15-minute intervals.

Let:

- S_f denote the scheduled arrival time of flight f,
- A_f denote the actual arrival time of flight f,
- $\delta_f = A_f - S_f$ denote the arrival delay,
- w_f denote the estimated walking time from gate to passport control,
- d_f denote the estimated disembarkation time.

Then the estimated passport-control arrival time of flight f is:

$$T_f = A_f + d_f + w_f \tag{2}$$

For each 15-minute interval t, the realized non-EU passenger demand is:

$$A_t = \sum_{f \in F_t} P_f \tag{3}$$

where F_t is the set of flights whose passengers are estimated to reach passport control during interval t.

4.3 Discrete-Event Simulation Layer

The fourth layer is a passenger-level discrete-event simulation implemented in `simpy`. In this layer, passengers arrive individually within each 15-minute interval and enter a queue for service at passport control. Border-control officers are modeled as service resources, and each passenger experiences a stochastic service time. The simulation tracks queue formation, waiting time, service completion, and queue carryover over time.

Let:

- τ_i denote the arrival time of passenger i to the queue,

- b_i denote the service start time of passenger i,
- c_i denote the service completion time of passenger i,
- s_i denote the service time of passenger i.

Then:

$$c_i = b_i + s_i \tag{4}$$

and the waiting time of passenger i is:

$$W_i = b_i - \tau_i \tag{5}$$

For each interval t, average waiting time is:

$$\bar{W}_t = \frac{1}{N_t} \sum_{i \in I_t} W_i \tag{6}$$

where:

- I_t is the set of passengers arriving in interval t,
- $N_t = |I_t|$ is the number of arrivals in interval t.

The queue length at the end of interval t is denoted by Q_t, and it emerges endogenously from the event-based interaction between arrivals and available service resources.

4.4 Staffing Policy Layer

The fifth layer defines the staffing rules to be tested in simulation. Three policies are evaluated.

Fixed staffing Under fixed staffing, the same number of officers is assigned in all intervals:

$$s_t = \bar{s} \tag{7}$$

where $\bar{s}$ is constant.

Reactive staffing Reactive staffing starts with fixed number of officers, and call new officers if queue is growing. In the operational implementation, any reinforcement beyond the currently available staff arrives after a response delay. I set it on 30 minutes.

Predictive dynamic staffing Predictive dynamic staffing uses forecast demand together with the current queue to assign officers proactively.

From a business perspective, the difference between the latter two policies is that reactive staffing responds after congestion becomes visible, whereas predictive dynamic staffing anticipates congestion using forecast information.

4.5 Performance Evaluation Layer

The final layer compares staffing strategies using operational key performance indicators. These include average waiting time, maximum queue length, average officers deployed, and average officer utilization.

5 Quantitative Analysis

The quantitative analysis in this project focuses on measuring how well the proposed framework improves passport-control operations for non-EU passenger flows. It is based on two numerical components: the predictive performance of the passenger-flow model and the operational performance of the staffing policies in the discrete-event simulation.

The first component is the demand-prediction model. The model estimates the number of non-EU passengers arriving at passport control in each 15-minute interval using flight-based and time-based explanatory variables. Its predictive performance is evaluated using standard regression measures,

namely mean absolute error (MAE), root mean squared error (RMSE), and the coefficient of determination (R^2). In this project, these measures are important because they indicate whether flight-based information can provide forecasts that are sufficiently accurate for staffing decisions.

The second component is the simulation-based comparison of staffing policies. After obtaining interval-level demand predictions, the project evaluates three staffing strategies in the `simpy`-based discrete-event simulation:

- fixed staffing,
- reactive staffing,
- predictive dynamic staffing.

For each policy, the simulation produces the following key performance indicators:

- average waiting time of passengers,
- maximum queue length,
- average number of officers deployed,
- and average officer utilization.

These measures are directly linked to the project objectives. Waiting time and queue length represent passenger service quality, while staffing levels and utilization represent operational efficiency. In this project, the quantitative comparison is used to determine whether prediction-based staffing performs better than static or delayed-response alternatives.

In addition, percentage improvements relative to a baseline can be used to summarize the impact of the proposed approach. If fixed staffing is taken as the benchmark, the improvement in average waiting time for another policy can be written as:

$$\text{Waiting time improvement} = \frac{W^{\text{fixed}} - W^{\text{policy}}}{W^{\text{fixed}}} \times 100 \tag{8}$$

where W^{fixed} is the average waiting time under fixed staffing and W^{policy} is the average waiting time under the alternative policy.

Similarly, the improvement in maximum queue length can be expressed as:

$$\text{Queue improvement} = \frac{Q_{\max}^{\text{fixed}} - Q_{\max}^{\text{policy}}}{Q_{\max}^{\text{fixed}}} \times 100 \tag{9}$$

Within the context of this project, the quantitative analysis therefore answers two practical questions: whether flight-based prediction improves staffing decisions, and how much operational benefit can be achieved in terms of lower waiting times and reduced congestion.

6 Results and Discussion

This section presents the main results of the project. The framework combines machine-learning-based prediction of non-EU passenger arrivals with a `simpy`-based discrete-event simulation in order to compare three staffing policies: fixed staffing, reactive staffing, and predictive dynamic staffing.

6.1 Demand Patterns

The demand analysis shows that non-EU passenger arrivals at passport control vary strongly over time. As shown in Figure 2 and Figure 3, passenger demand changes from day to day and is also uneven across 15-minute intervals. This means that passport-control demand is not stable and cannot be managed effectively with a single constant staffing level.

This result is important because queue growth depends not only on the total number of passengers in a day, but also on how many passengers arrive at the same time. Even if total daily traffic is moderate, large short-term peaks can still create long queues.

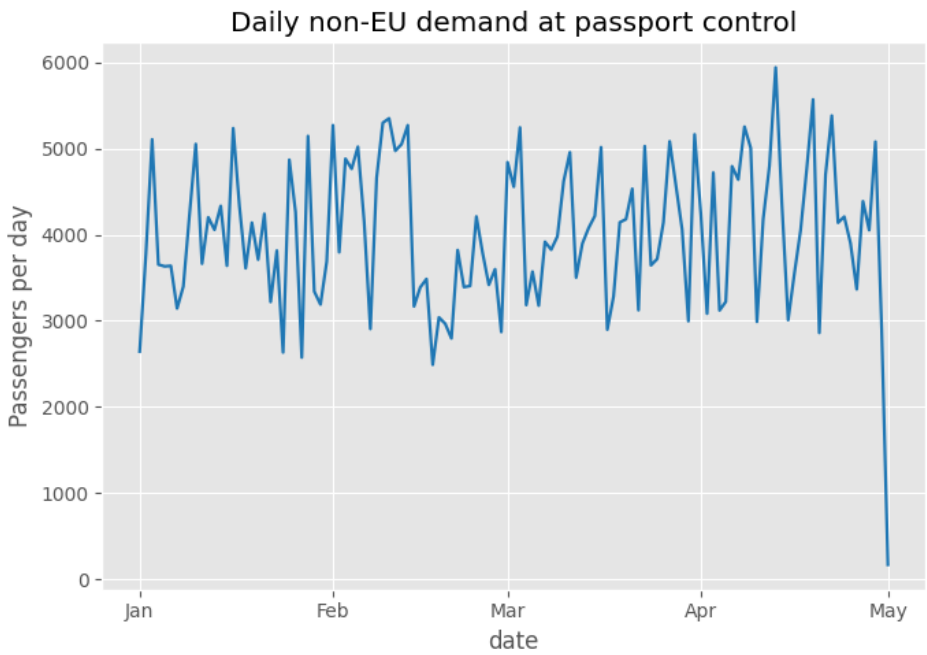

Figure 2: Daily non-EU passenger demand.

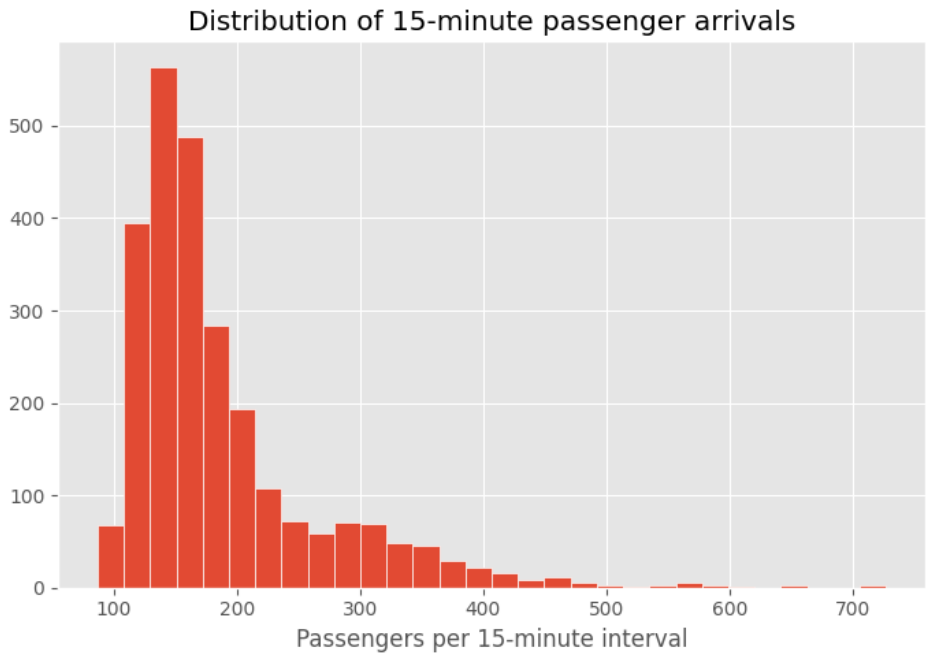

Figure 3: Distribution of non-EU passenger arrivals in 15-minute intervals.

6.2 Prediction Results

The machine-learning model was trained to predict the number of non-EU passengers arriving at passport control in each 15-minute interval. Figure 4 shows the comparison between predicted and actual arrivals.

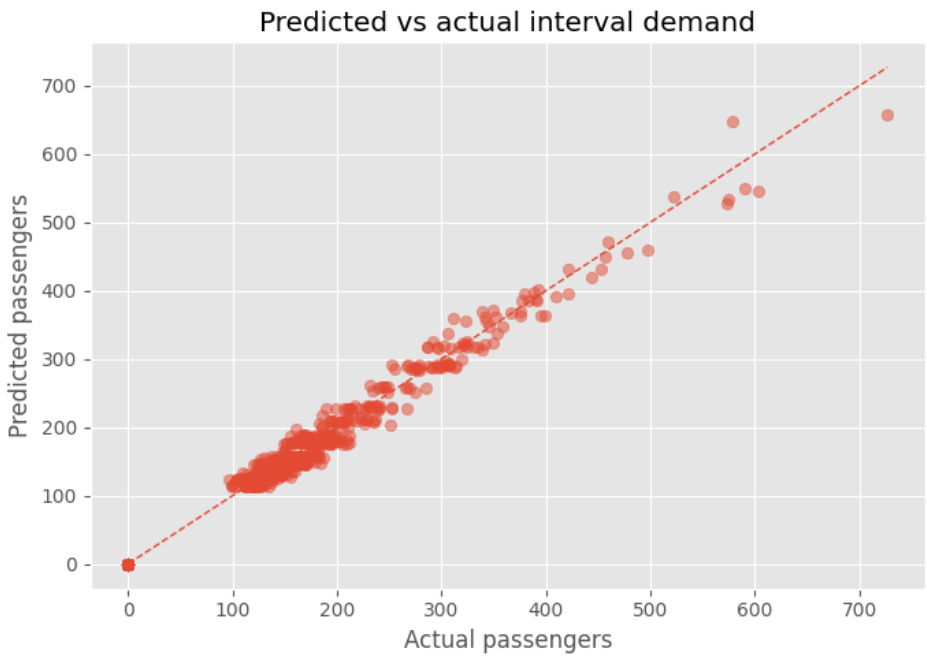

Figure 4: Prediction diagnostics for non-EU passenger arrivals at passport control.

The model captures the main changes in passenger demand and identifies the most important peaks and low-demand periods. Although the predictions are not perfect, they are useful for staffing decisions because they provide early information about when congestion is likely to occur.

In this project, the value of prediction is practical rather than purely statistical. The goal is not only to predict exact passenger numbers, but to give airport managers enough information to prepare staffing before queues become too long.

6.3 Simulation Results

Passenger processing at passport control was modeled using a `simpy`-based discrete-event simulation. In this simulation, passengers arrive individually, wait in a queue, and are processed by available officers. This makes the queue behavior more realistic than a simple interval-based capacity model.

Three staffing policies were tested:

- **Fixed staffing:** the same number of officers is used throughout the day.
- **Reactive staffing:** staffing follows a planned baseline and requests extra officers when the queue becomes too long, but these officers need time to arrive.
- **Predictive dynamic staffing:** staffing is adjusted using forecast passenger demand and current queue conditions.

Table 1 summarizes the main results. The values in the table should be updated with the latest notebook outputs.

The fixed staffing policy performs worst when demand increases sharply, because it cannot adapt to changing passenger flows. This leads to long queues and higher waiting times.

The reactive staffing policy performs better than fixed staffing because it can request additional officers when the queue grows. However, it is still limited and requires more officers on the ground

Table 1: Performance Comparison of Staffing Policies

Policy	Avg Wait (min)	Max Queue (pax)	Avg Officers	Utilization
Predictive Dynamic	6.83	850	4.46	0.23
Reactive Staffing	7.52	965	6.00	0.26
Fixed Staffing	23.44	1646	4.00	0.38

because reinforcement is delayed. In practice, this means the queue may continue to grow before the extra officers actually arrive.

The predictive dynamic staffing policy performs best overall. Because it uses forecast demand, it can increase staffing earlier and reduce the size of queue peaks before they become severe. This leads to lower waiting times and better queue control.

6.4 Operational Meaning of the Results

The results show that queues at passport control are mainly caused by short-term mismatch between arriving passengers and available officers. If staffing is too rigid, even a short arrival peak can create delays that continue into later periods.

Reactive staffing improves the situation because it allows some response to congestion, but it still acts after the problem is visible. Predictive dynamic staffing is more effective because it acts earlier, using forecast information. This makes it a more proactive strategy.

From a business perspective, this means that better staffing decisions do not always require many more officers. In many cases, they require better timing of officer allocation. This is an important result for airport management, where staffing resources are often limited.

6.5 Discussion

Overall, the results show that flight information can be used to improve staffing decisions at passport control. The demand analysis confirms that non-EU passenger arrivals are uneven over time. The prediction model shows that these changes can be forecast in advance. The simulation then shows that using these forecasts in staffing decisions can reduce queues and waiting times.

The main conclusion is that passport-control staffing should not be treated as a fixed daily schedule only. It should be managed as a dynamic process that responds to changing passenger demand. In this project, predictive dynamic staffing provides the best overall performance.

6.6 Future Development

Although the proposed framework demonstrates the value of combining prediction and simulation for passport-control staffing, it remains a simplified decision-support model. Several important business and operational factors are not yet included and should be considered in future development.

For example, the current model does not include staff shift structures. In real airport operations, officers work according to fixed shift patterns, break periods, and handover times. This means that staffing cannot always be adjusted freely from one interval to the next. Future versions of the model may include shift schedules and the practical limits they impose on staffing flexibility.

References

[Kha23] Mutahhir Khan. Airport queue simulation report, 2023.

[mdp19] A queueing model for airport operations. *Aerospace*, 2019.